Written Report:
Manager Performance and Workload Analysis

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Introduction

The purpose of this report is to analyze manager performance and workload analysis using data from an HR company. This will help provide insight into the performance and working conditions of the company's managers. This insight will be provided by analyzing manager data, visualization, and Q&As to support strategic HR decisions. Additionally, in order to complete this analysis, three datasets will be used: managers.csv, managers_append.csv, and manager_bonus.csv. The managers.csv dataset is the main dataset. It contains information about each manager, such as, performance group, years employed, test scores, group size, city, etc. The managers_append.csv dataset contains additional records for the managers that were missing in the main dataset. Lastly, that manager_bonus.csv dataset contains bonus information that was collected for most managers. This includes annual bonus scores and review years.

Data Preparation

In order to begin the analysis, the first step is to prepare the manager's (managers.csv) dataset. As mentioned earlier this is the main dataset. The managers.csv and managers_append.csv dataset were appended into a new variable. This is to ensure that the manager's dataset is complete. The next step was merging the manager_bonus.csv with the appended dataset using the variable employee_id. All the managers from the main dataset were retained, and that is true even if the bonus information was unavailable for some of the employees. This way the combined dataset will contain all the manager characteristics, performance information, and bonus scores.

Summary of Manager Characteristics

In order to understand the manager population, two numerical variables and two categorical variables were calculated. The two numerical variables that were tested are yrs_employed and test_score. The two categorical variables are performance_group and city. Here is what the findings said:

Numerical Variables:	**Categorical Variables:**
yrs_employed • Most managers have at least 4 years or more of managing experience	**performance_group** • 13% of managers were placed at the top

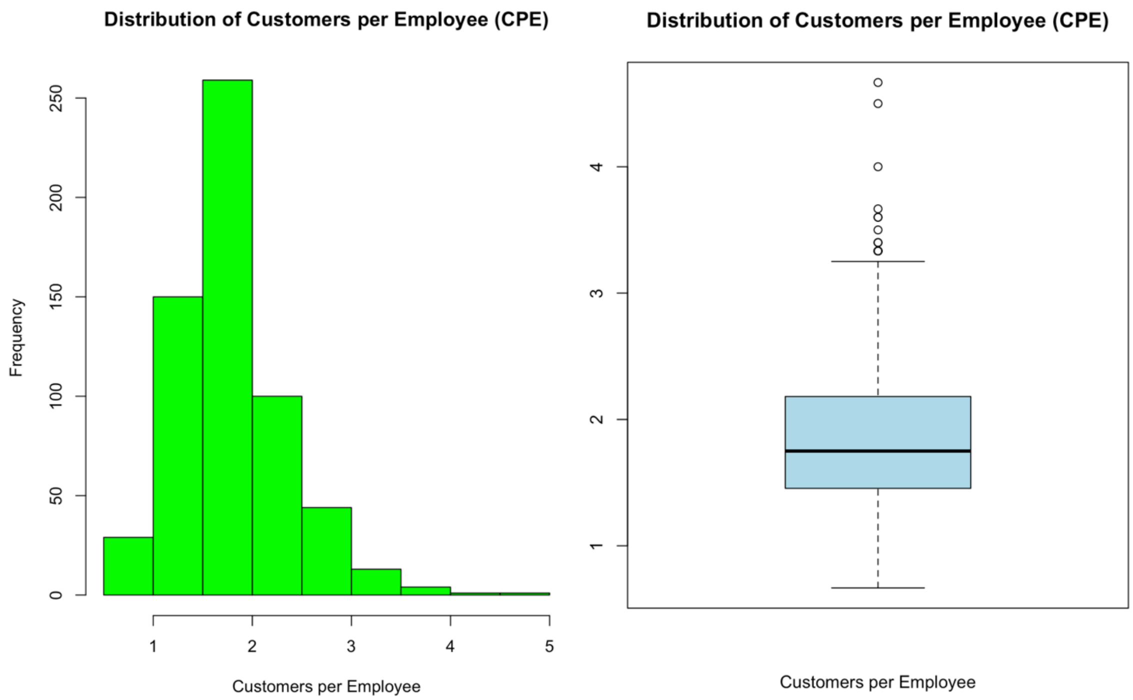
	• 64% of managers were placed in the middle • 23% of the managers were placed at the bottom
test_score • Most managers scored an average score of 242	**city** • 11% of the managers are located in Chicago • 4% of the managers are located in Houston • 1% of the managers are located in Miami • 34% of the managers are located in New York • 5% of the managers are located in Orlando • 10% of the managers are located in San Francisco • 35% of the managers are located in Toronto.

The results suggest that a large portion of managers fall into the middle performance group. Since there is a large number of managers in that middle range the organization should help them move into the top performance group. Additionally, help the bottom performing managers move up to the middle performance group, and then eventually help them move to the top performance group. Also, most managers are located specifically in New York and Toronto. That is their main concentration area for manager location. With Chicago and San Francisco being in the middle and Miami, Houston, and Orlando being the lowest. Most of these managers had at least 4 years or more of managing experience, with an average test score of 242.

Customers per Employee (CPE)

In order to measure the efficiency of the manager's workload, a new variable called CPE (Customer per Employee) was created. In order to create this variable:

CPE = customers / group_size



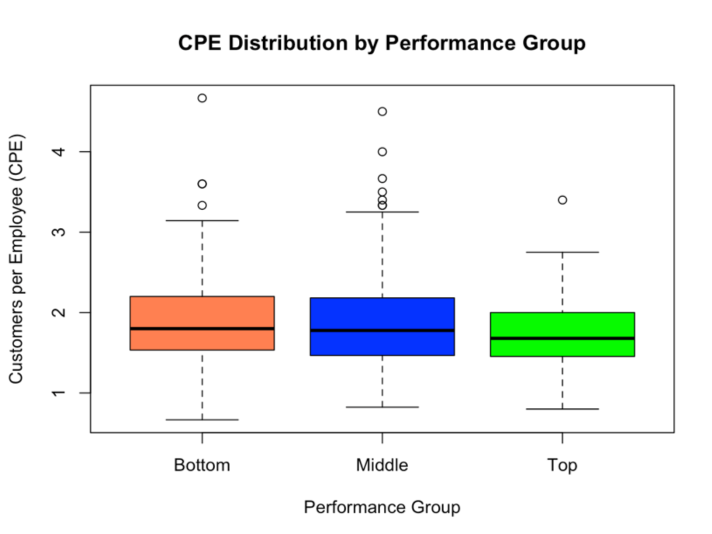
This will represent how many customers are handled per employee. The results show that most of the managers have a moderate number of customers. However, there are several outliers. 14 outliers to be exact were identified. These outliers do indicate that there were some managers with a significantly higher workload compared to the other managers.

The results also show that managers with more than 3 customers per employee have a higher demand. Specifically 19 managers are classified as high-demand managers.

These results were then subset into a new variable, containing the following variables: employee_id, city, performance_group, and CPE>3. Then the CSV for this new created variable was saved for future usage to analyse and interpret more data.

Performance Group and Efficiency

In order to analyze the performance group and efficiency, a boxplot was created. This boxplot compares the distribution of CPE (customers per employee) across performance groups. Afterwards, the average CPE for each performance group was calculated.



Performance Group	Average CPE
Top	1.74
Middle	1.85
Bottom	1.92

The results show that the top-performing managers did not serve the most customers. On the contrary, the bottom performance group managers had the highest CPE rate, followed by middle, and last but not least, top performing managers.

These results also suggest that groups that have a higher workload intensity can be associated with lower performance ratings. This is because having an excessive workload will reduce the level of effectiveness.

Bonus Score Analysis

There were 54 managers missing their bonus scores. The average score of the bonuses was calculated and came out to 73.80, and in order to make sure the dataset was complete, all the missing bonus values were set to the average (73.80). This way, the missing data will not cause any issues when analyzing future data.

Performance and Long Working Hours

To examine the relationship between the performance group and the long work hours during the previous week, a contingency table was created to find out whether higher-performing managers tend to work long hours, and if high-hour workload is independent of performance.

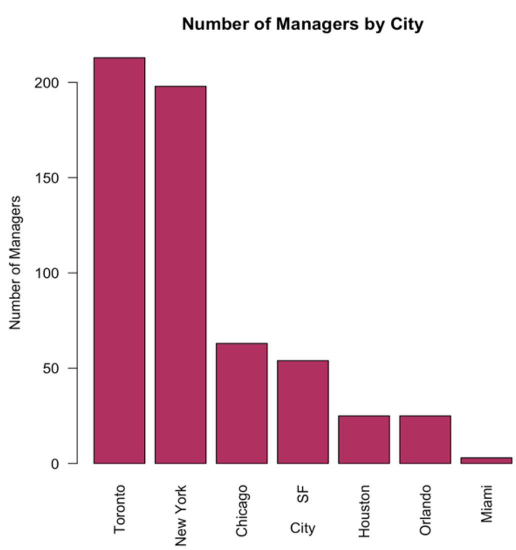
Performance Group	Percentage of Long Hours Works

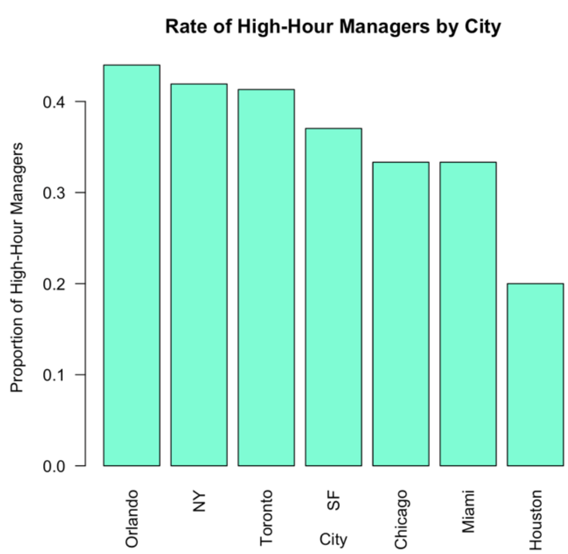
Top	55%
Middle	39%
Bottom	30%

Based on this data, it is shown that, yes, high-performing managers do tend to work longer hours. Afterwards, a chi-square test was conducted. This was to determine if the relationship between these two was statistically significant. The p-value was 0.0006497, and if the p-value<0.05, then the null hypothesis: high-hour workload is independent of performance, is rejected. Meaning that a higher-hour workload is not independent of performance, and that, in fact, higher-performing managers are more likely to work long hours compared to other performance groups.

City Analysis

In order to do a city analysis on the managers, two bar plots were created: Managers by City & High-Hour Manager Rate by City.





The results show that the city with the largest number of managers is Toronto and the smallest number of managers is in Miami. For HR, this can help them see where their main focus should be on retaining and training managers to be top performers. For the rate of high-hour workload, the results show that a portion of managers are actually working long hours regardless of the city they are located in. This can be because, depending on the city and location, there can be different amounts of workload needed along with the expectations. This can be due to local markets, staffing, structure, economy, etc.

Test Score and Experience

In order to analyze the relationship between test scores and years employed for managers, a scatterplot was created, and then the correlation coefficient was calculated.

The results of the data show that the r value is 0.0098, which is very close to 0. This indicates that there is no meaningful linear relationship between experience and test scores for managers, as shown in the graph above. This indicated to HR that high test scores are not a result of years of experience or the lack thereof.

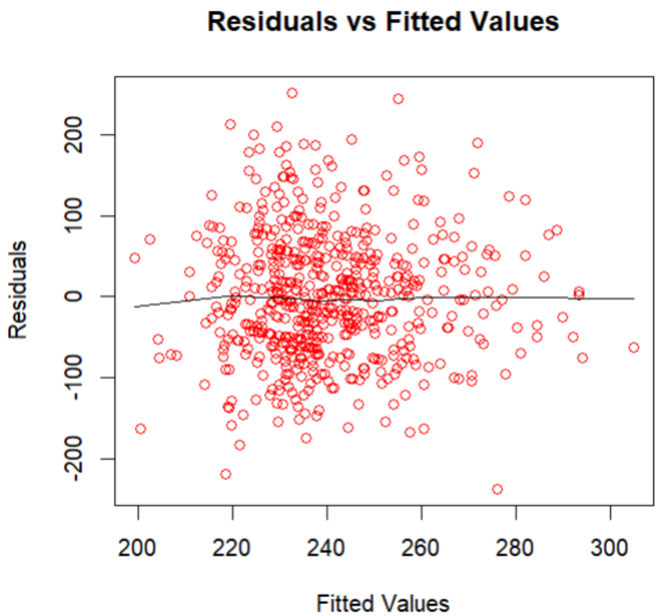
Linear Regression: Predicting Test Scores

In order to predict test scores a linear regression model was created. The estimates used to predict test scores were yrs_employed, group_size, and customers. Using this linear regression model, it gave regression coefficients related to the three used variables. The intercept (201.27) shows the predicted test score when yrs_employed, customers, and group_size are all equal to zero. It is the general baseline for the regression model.

The coefficient for yrs_employed is -3.664, which shows that there is a 3.66 point decrease in the predicted test score for every year of employment. This is not significant because the P value is ($P = 0.352$), which is higher than 0.05 for us to consider it statistically significant. The coefficient for customers is -0.532, which shows there is a 0.53 point decrease in predicted test score for each additional customer added to an employee. This is not significant because the P value is ($P = 0.436$), which is higher than 0.05 for us to consider it statistically significant. The coefficient for group size is 5.661, which shows that each additional person in a group is associated with a 5.66 point increase in the predicted test score. This is significant because the P value is ($P < 0.001$), which is lower than 0.05, which makes it statistically significant.

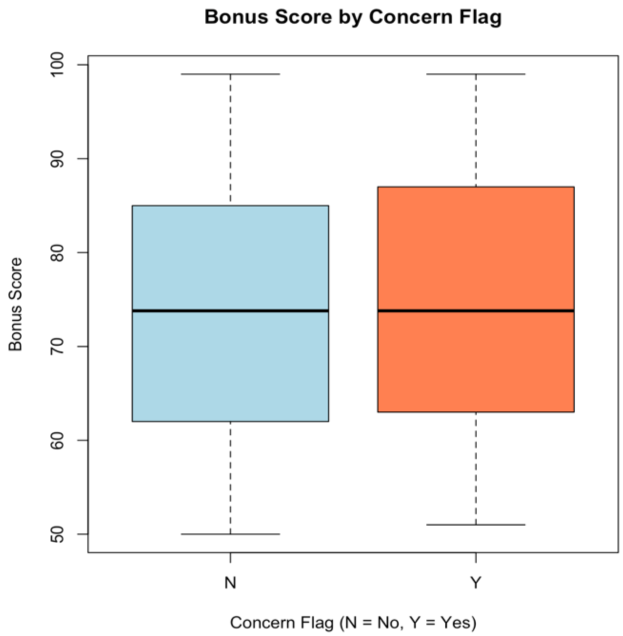
We are then asked to predict the expected test score for a manager with 10 years of employment, 20 group size, and 25 customers. This can be done with a prediction model in R using a data frame that puts in the values the model will use to make its prediction. It is shown that the expected test score for a manager with 10 years of employment, 20 group size, and 25 customers is 264.5452.

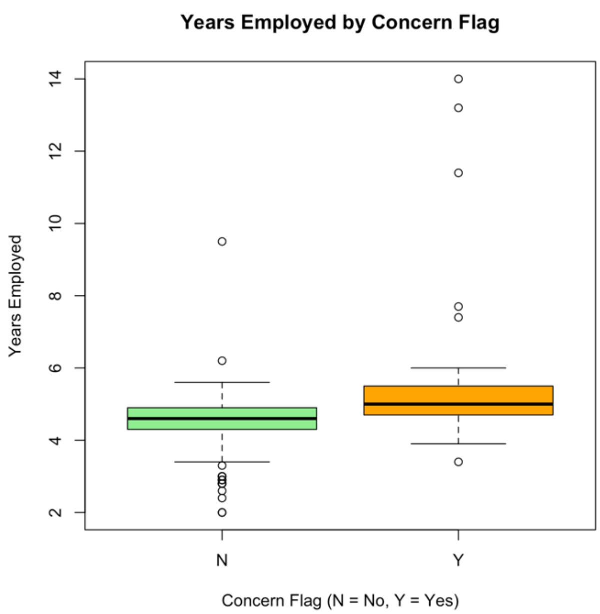
A residual plot is needed to determine whether there is any relation between the residual values and the fitted values. This can be done using the scatter.smooth plotting model, which will display any patterns found through the values.

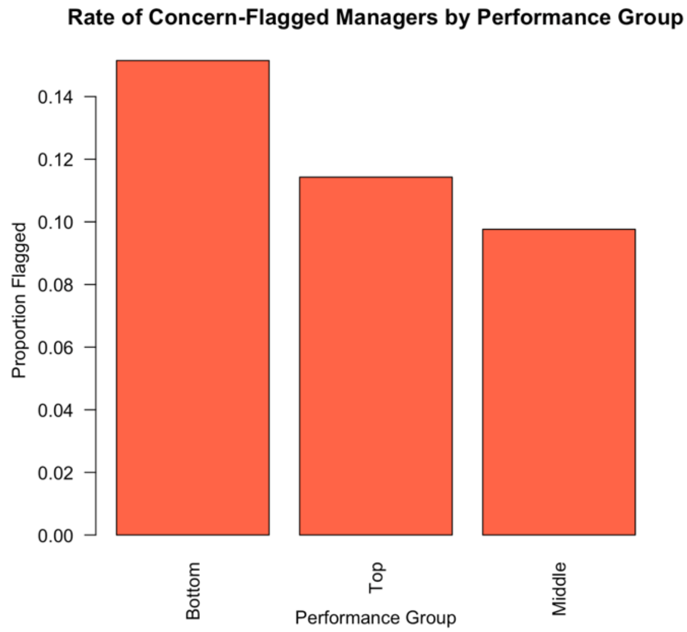


From the plot, it is seen that there is a general absence of pattern and no apparent curvature. This shows that there isn't much of a relationship with test scores, years employed, and group size.

Additional Analysis: Concern Flags


Bonus Score by Concern Flag
Bonus Score
100
90
80
70
60
50
N
Y
Concern Flag (N = No, Y = Yes)


Years Employed by Concern Flag
Years Employed
14
12
10
8
6
4
2
N
Y
Concern Flag (N = No, Y = Yes)



For this part of the analysis, we looked at the managers who had a concern flag and compared them with managers who did not have one. The goal was to see if there were any differences between these two groups. Firstly, we looked at the bonus scores, where the results showed that managers with a concern flag usually had slightly lower bonus scores compared to managers without a flag, which might mean that managers who are flagged are not performing as well as the others.

Next, we looked at the number of years the managers have worked at the company. Managers with a concern flag generally had fewer years of experience, which could mean that newer managers are more likely to be flagged because they are still learning how things work in the company. In addition, we also looked at the performance groups.

The results showed that the bottom performance group had the highest number of concern flags, while the top performance group had the lowest number. This shows that managers with lower performance are more likely to have concerns. Overall, this analysis suggests that HR should pay attention to managers who are flagged for concern. These managers may benefit from extra training, support, or mentoring to help them improve their performance.

Conclusion

This analysis of the manager's performance and data reveals many flaws and important patterns that the Human Resources department must address. One thing that prevailed from the analysis was that the majority of managers were categorized within the "middle" performance group (64% of them). This presents a clear opportunity for the program to focus on helping this group of managers advance so they can get treated how they deserve and how they can have increased growth through developmental programs. Another thing that needs attention is how imbalanced the workload is for the "bottom" managers since they have the lowest level of performance, but have the highest number of customers per employee, which makes it all the more necessary to assess.

Data also ended up revealing that, on average, the “top” performing managers work longer hours, and that raises the question as to how it is that they do not end up burning out as the “bottom” performers do. The newer and less experienced managers are more likely to get flagged, and once managers end up being flagged, they have lower bonus scores and most of the time end up entering the “bottom” group. This goes to show that things like mentorship, regular training, and meetings with other managers would be nothing but beneficial for those who are in their infant stages as managers in the company.

To round things out, upon analysis, it ended up becoming clear that group size was the main factor that contributed to test scores, and although it seemed as if experience would’ve led the way towards success, that was not the case since staffing and team structure decisions played a huge role in the outcomes of the managers. Overall, the results of this group analysis revealed how having analysts support decisions through data, such as hiring, training, and team structure, can truly help an HR department be the backbone of an effective team.